

Supervised machine learning

DATA SCIENCE FOR BUSINESS



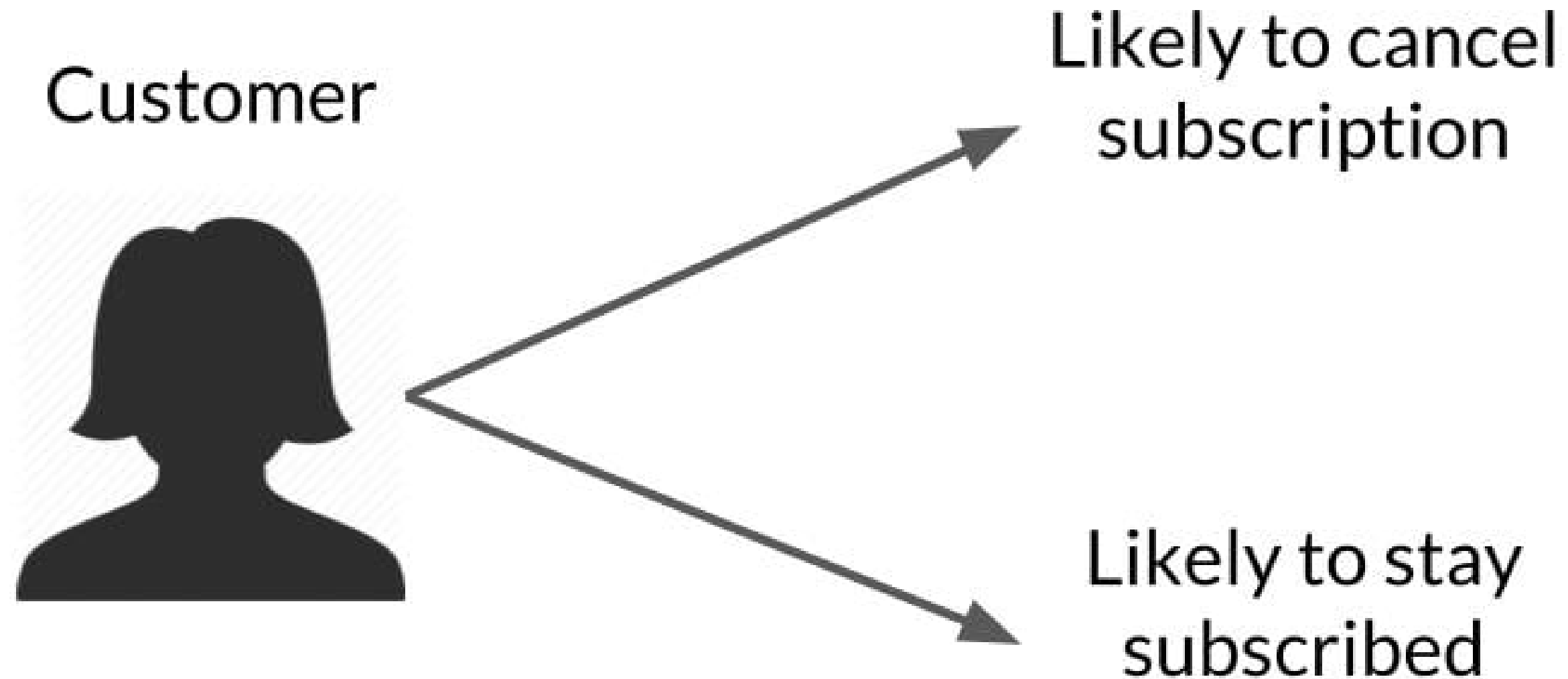
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What is supervised machine learning?

- Machine learning: Predictions from data
- ***Supervised machine learning***: Predictions from data with *labels* and *features*
 - Recommendation systems
 - Email subject optimization
 - Churn prediction

Case study: churn prediction



Case study: churn prediction

Training
Data:
Customers



Case study: churn prediction



Labels
Customer
outcomes

churn
subscribe
subscribe
churn
subscribe
churn

Case study: churn prediction

	Age	Gender	Date of last purchase?	Date of last visit?	Likes cats?	Household \$\$	Location	Number of Kids	Profession	
										churn
										subscribe
										subscribe
										churn
										subscribe
										churn

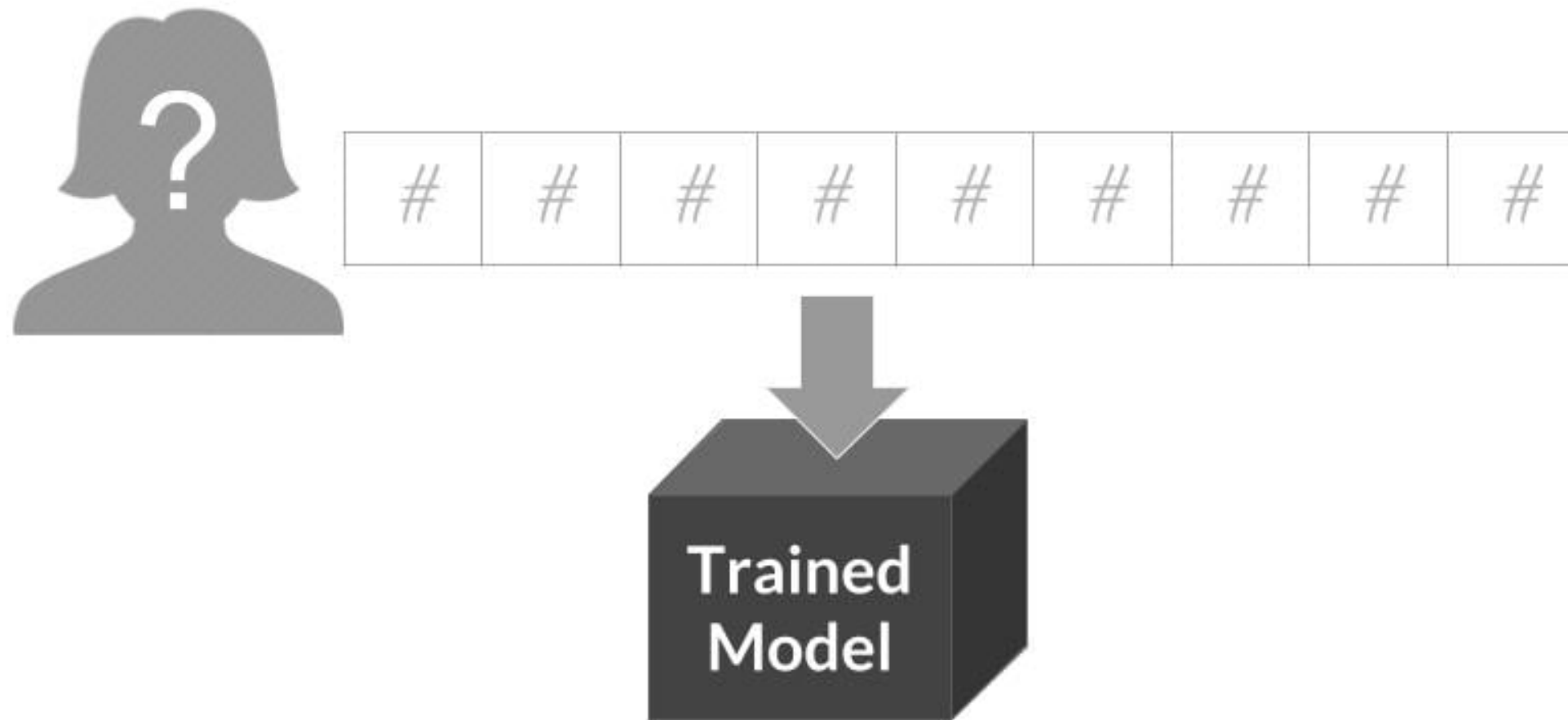
Features:
Collected customer data

Case study: churn prediction

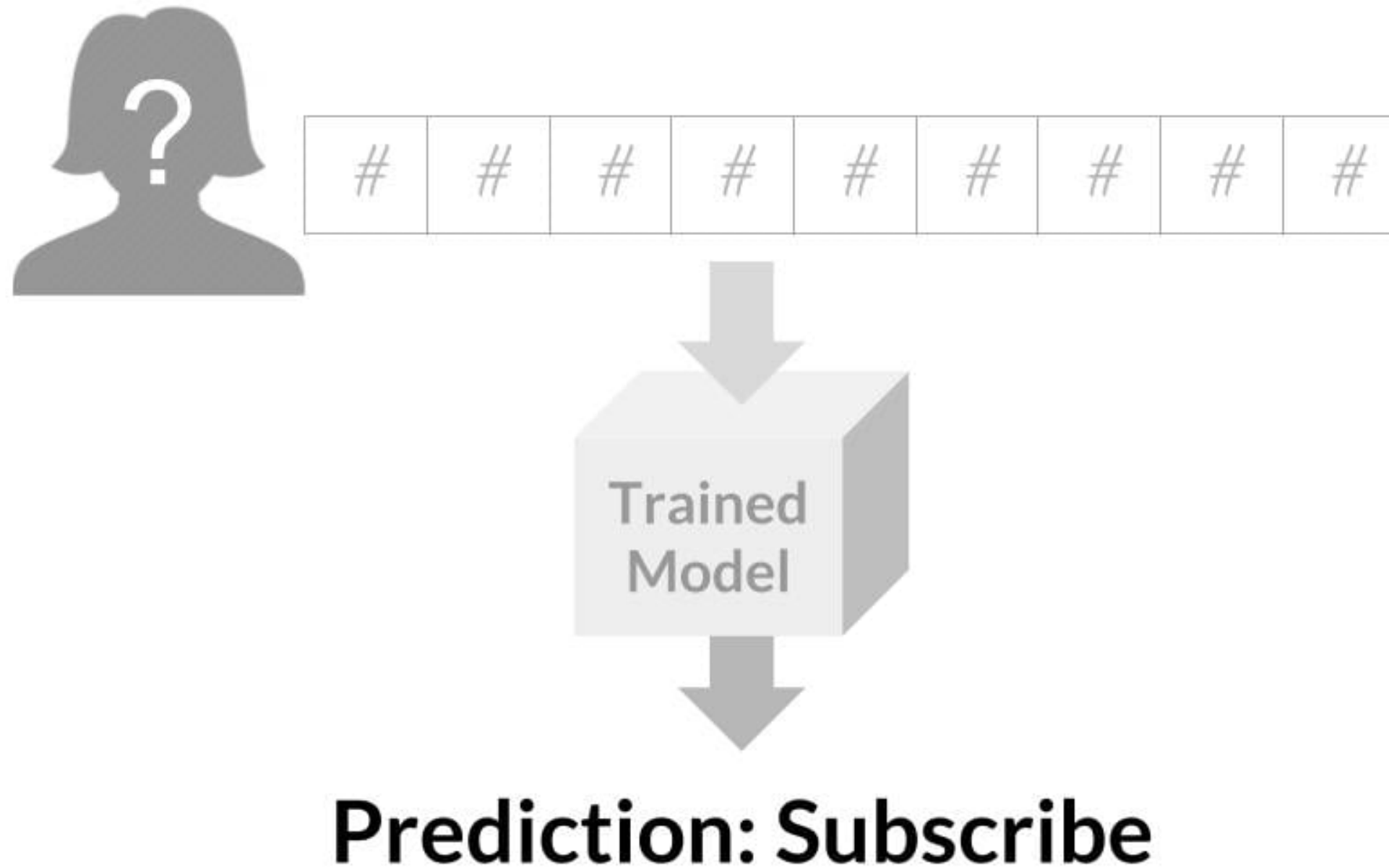


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Case study: churn prediction



Case study: churn prediction



Recap

- Make a prediction based on data
- Data has *features* and *labels*
 - Label: what we want to predict
 - Features: data that might predict the label
- Trained model can make predictions

Model evaluation

Split historical data into training and testing sets



Model evaluation

	Prediction	Reality
Churn	0%	3%
Remain	100%	97%

Let's practice!

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Clustering

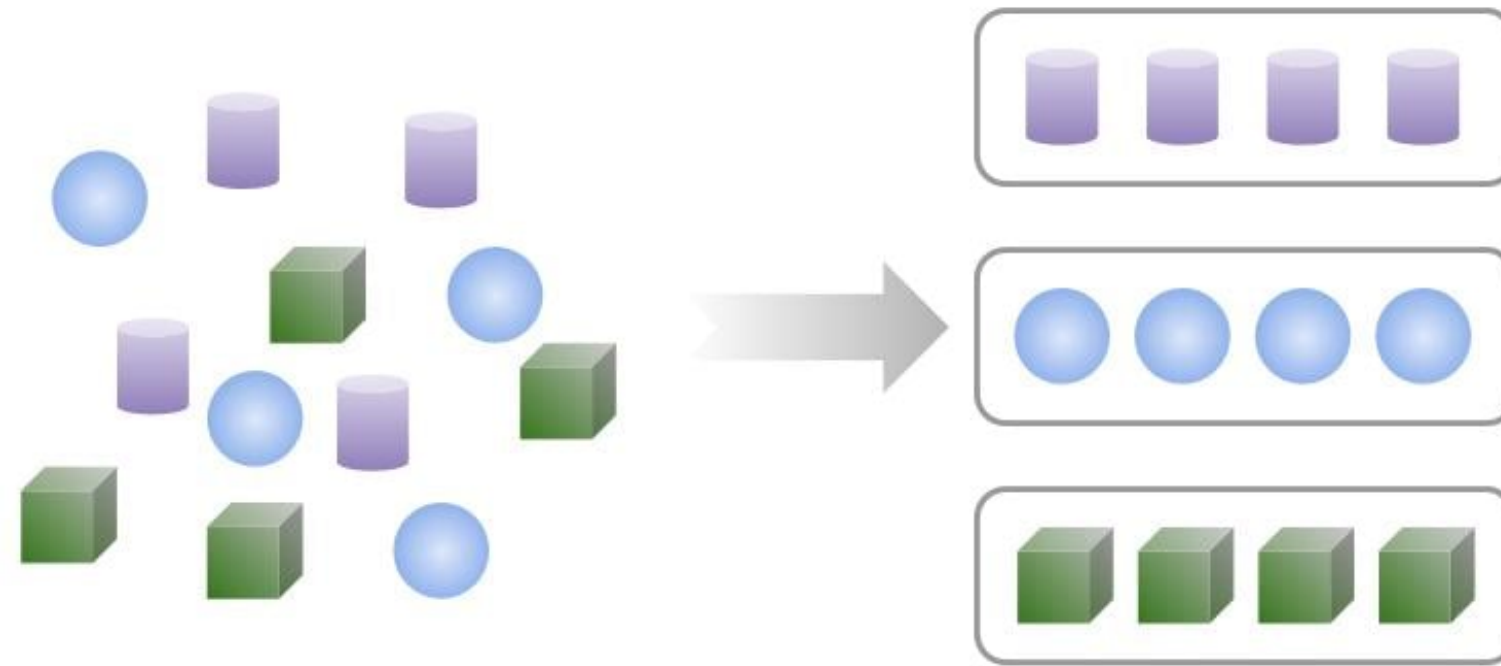
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What is clustering?



- Divide data into categories
- Use cases
 - Customer segmentation
 - Image segmentation
 - Anomaly detection

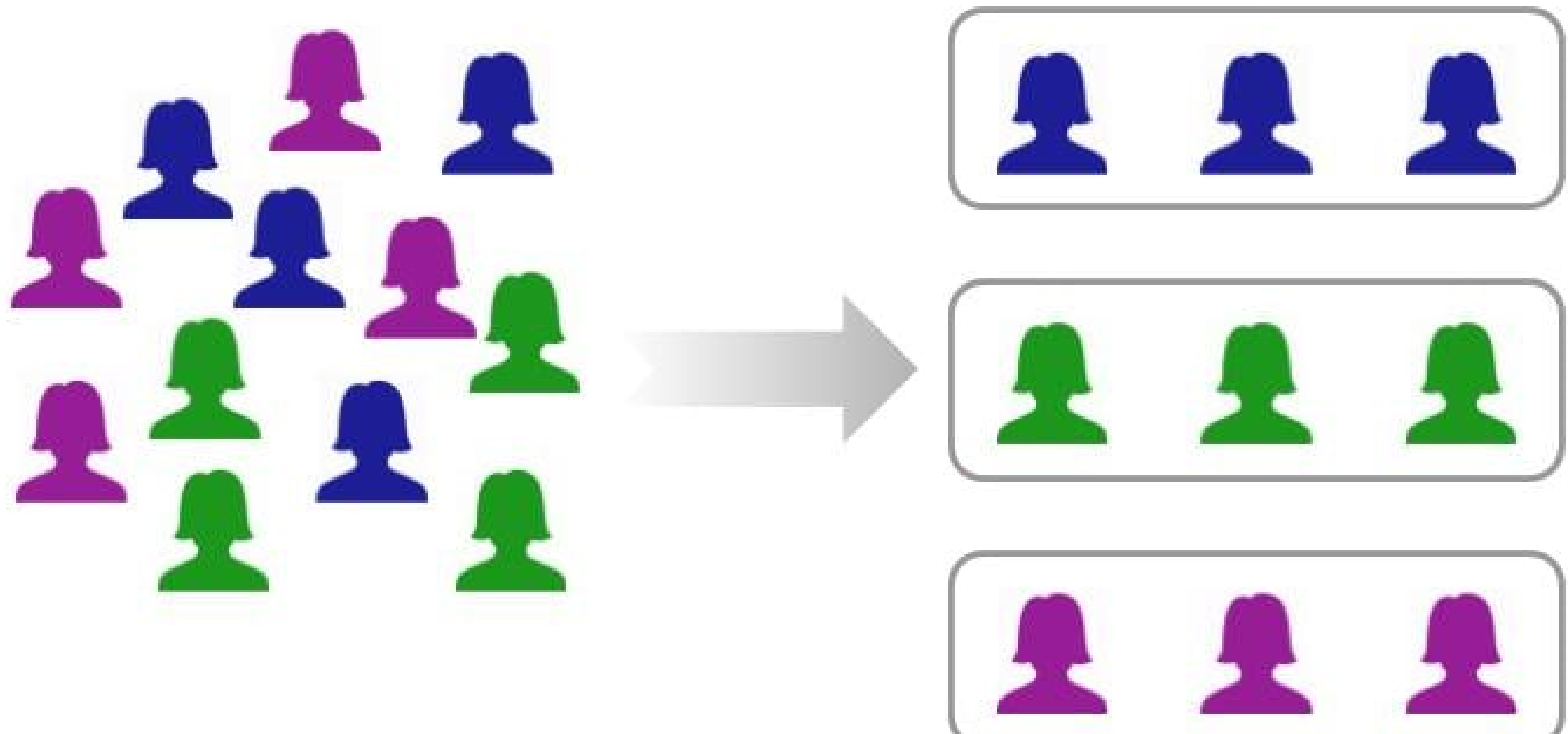
Supervised Machine Learning

	Age	Gender	Date of last purchase?	Date of last visit?	Likes cats?	Household \$\$	Location	Number of Kids	Profession	
										churn
										subscribe
										subscribe
										churn
										subscribe
										churn

Unsupervised Machine Learning

	Age	Gender	Date of last purchase?	Date of last visit?	Likes cats?	Household \$\$	Location	Number of Kids	Profession
									
									
									
									
									
									

Case study: customer segmentation



Case study: customer segmentation

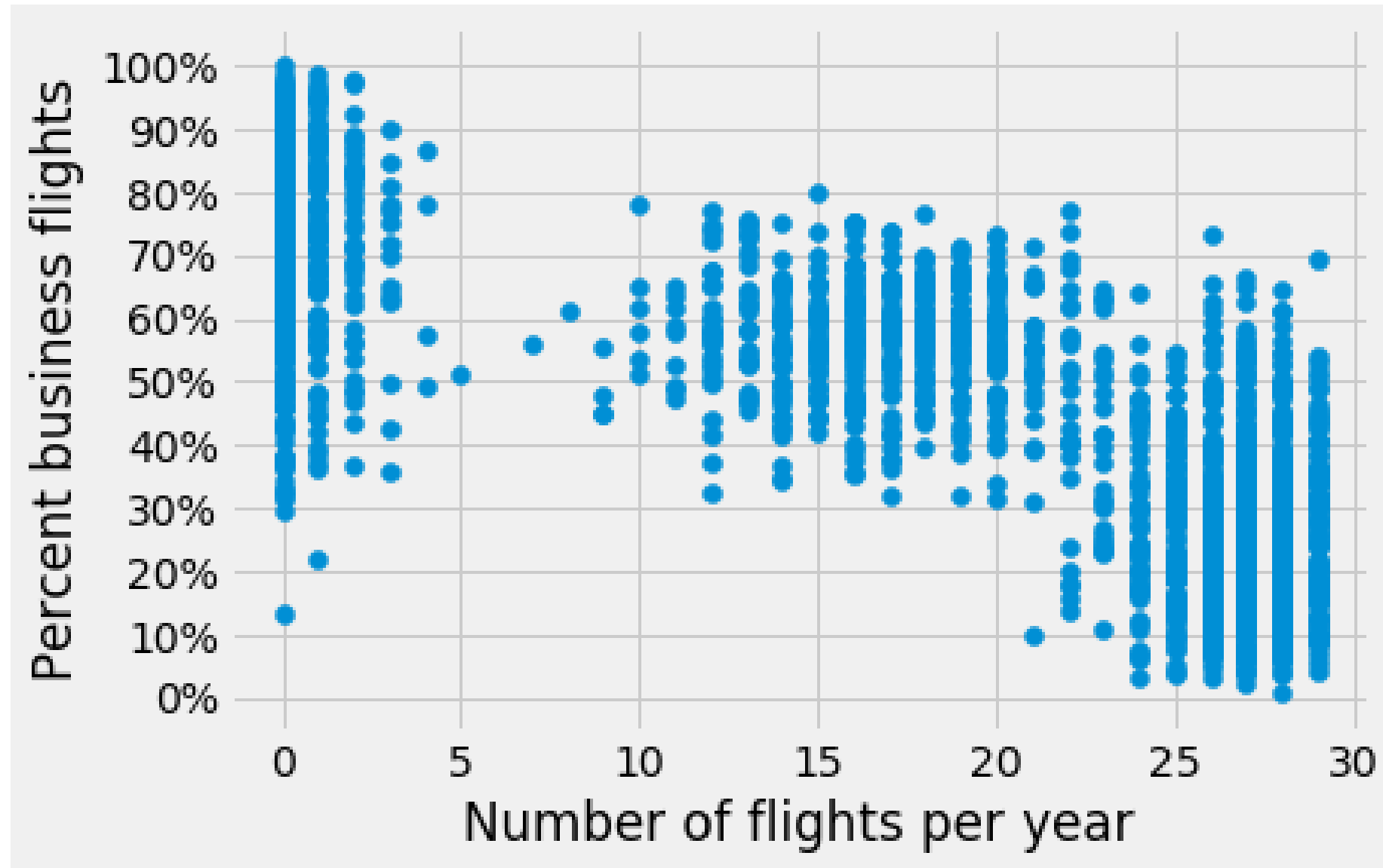
Define features

- Number of flights in the past year
- Percent international
- Advanced planning
- Percent business class

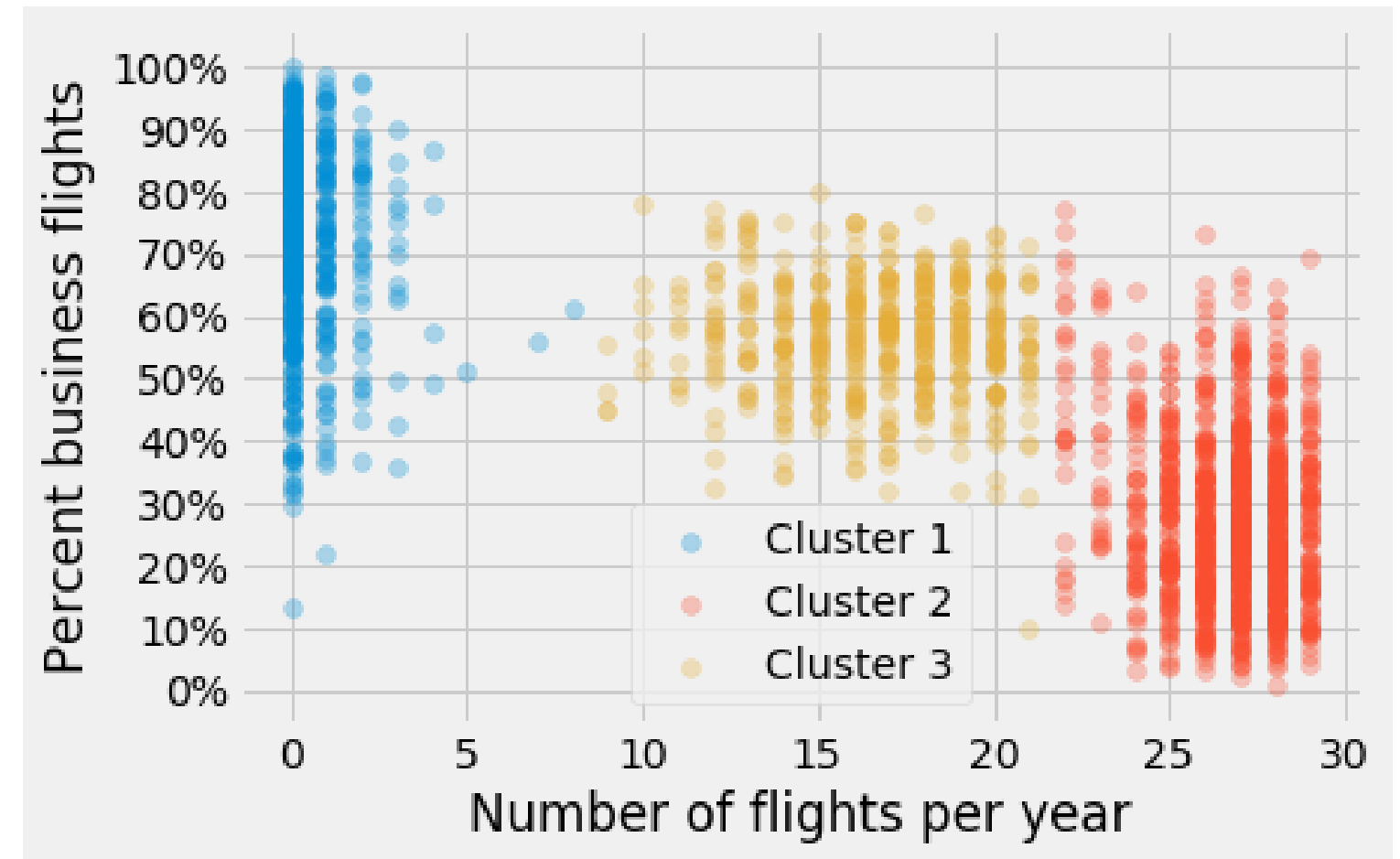
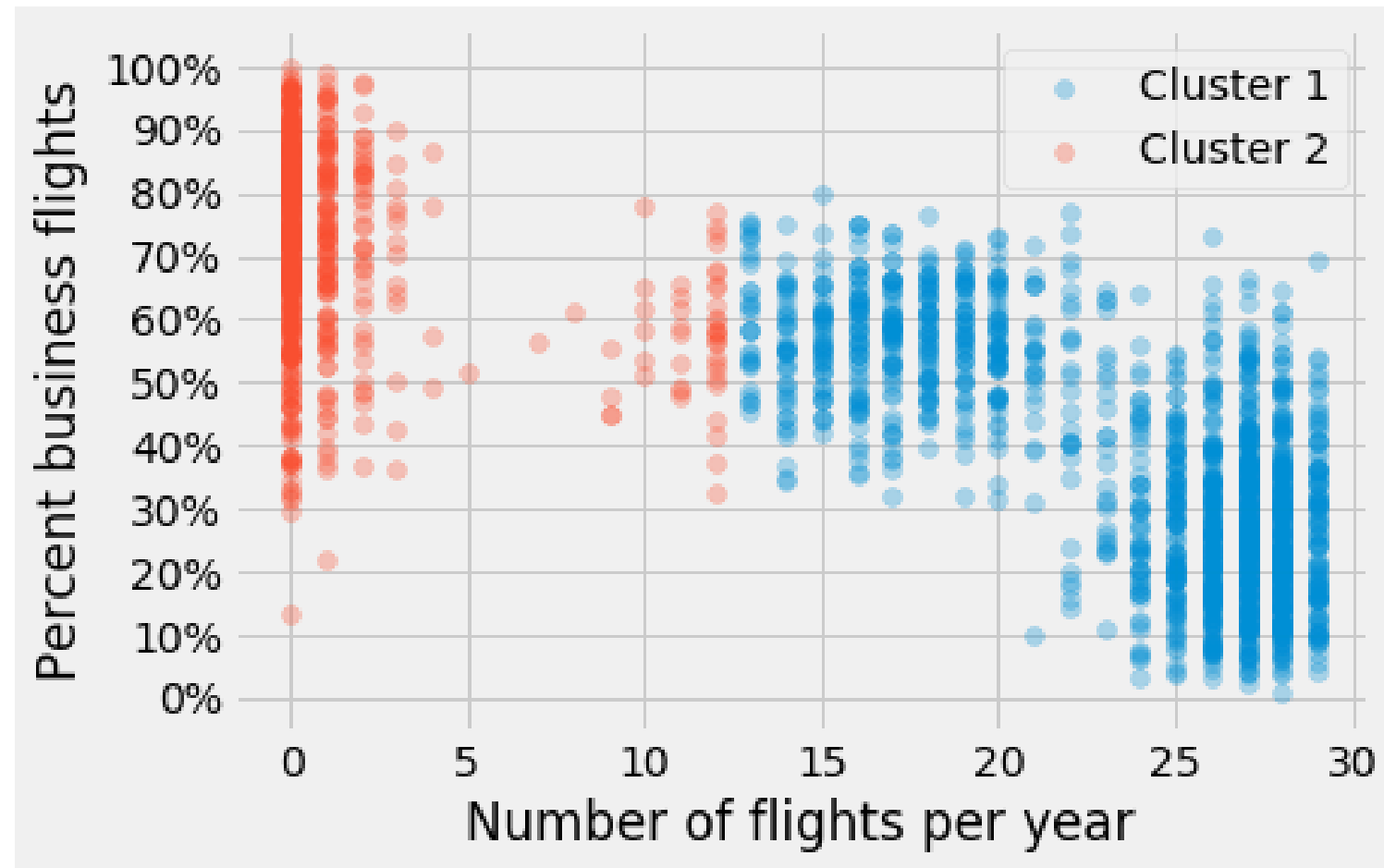


Case study: customer segmentation

- Define number of clusters



Case study: customer segmentation



Clustering review

Definition

- Divide unlabeled dataset into different categories

Steps

- Select features
- Select number of clusters
- Use clusters to solve business problems

Let's practice!

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Special topics in Machine Learning

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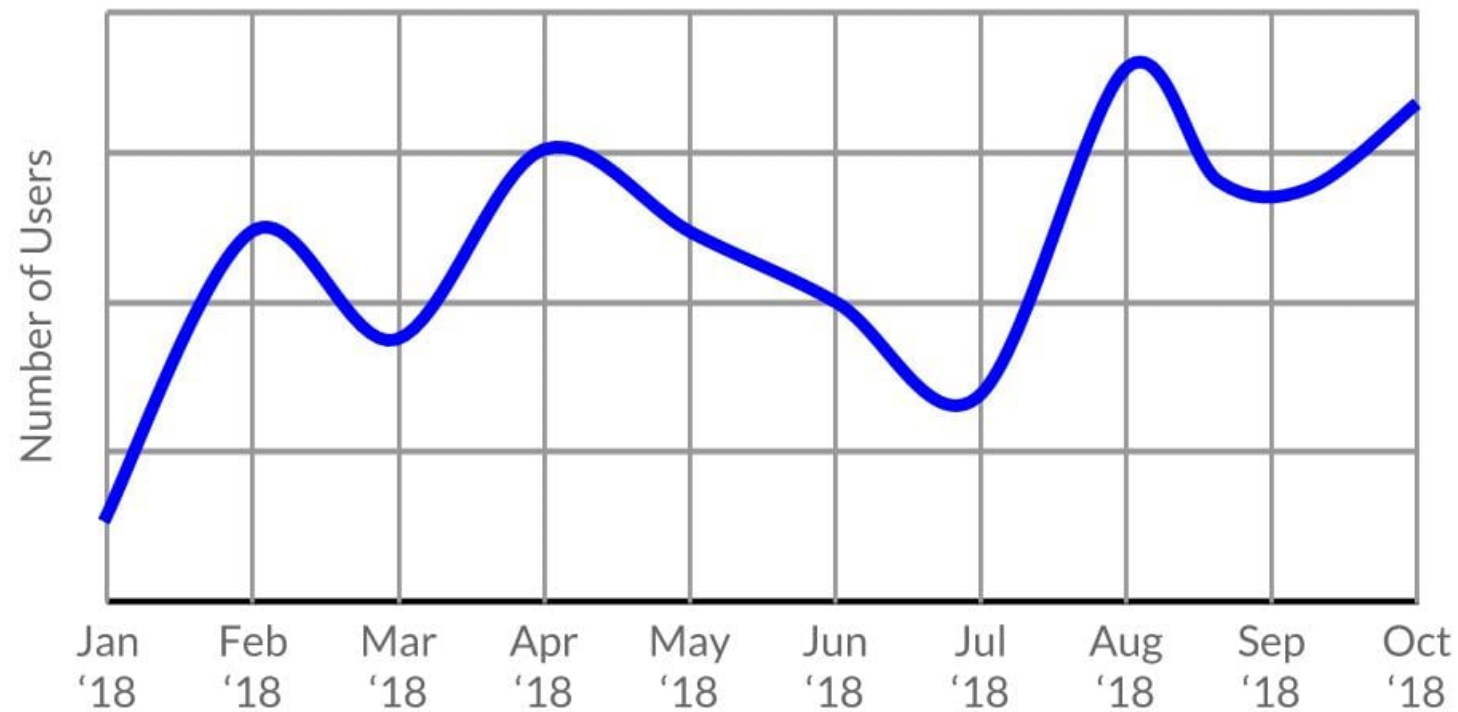


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Time series forecasting

Monthly Active Users



- Time is a feature
- Accounts for weekly, monthly, or yearly trends

Seasonality

- **Weekly:** Lower television viewership on Fridays
- **Monthly:** Higher spending at end of pay periods
- **Yearly:** Less ice cream in the winter



Natural Language Processing

- Dataset is text
 - Customer reviews
 - Tweets
 - Medical records
 - Email subjects
- Possible uses
 - Classifying sentiment
 - Clustering medical records



Word counts

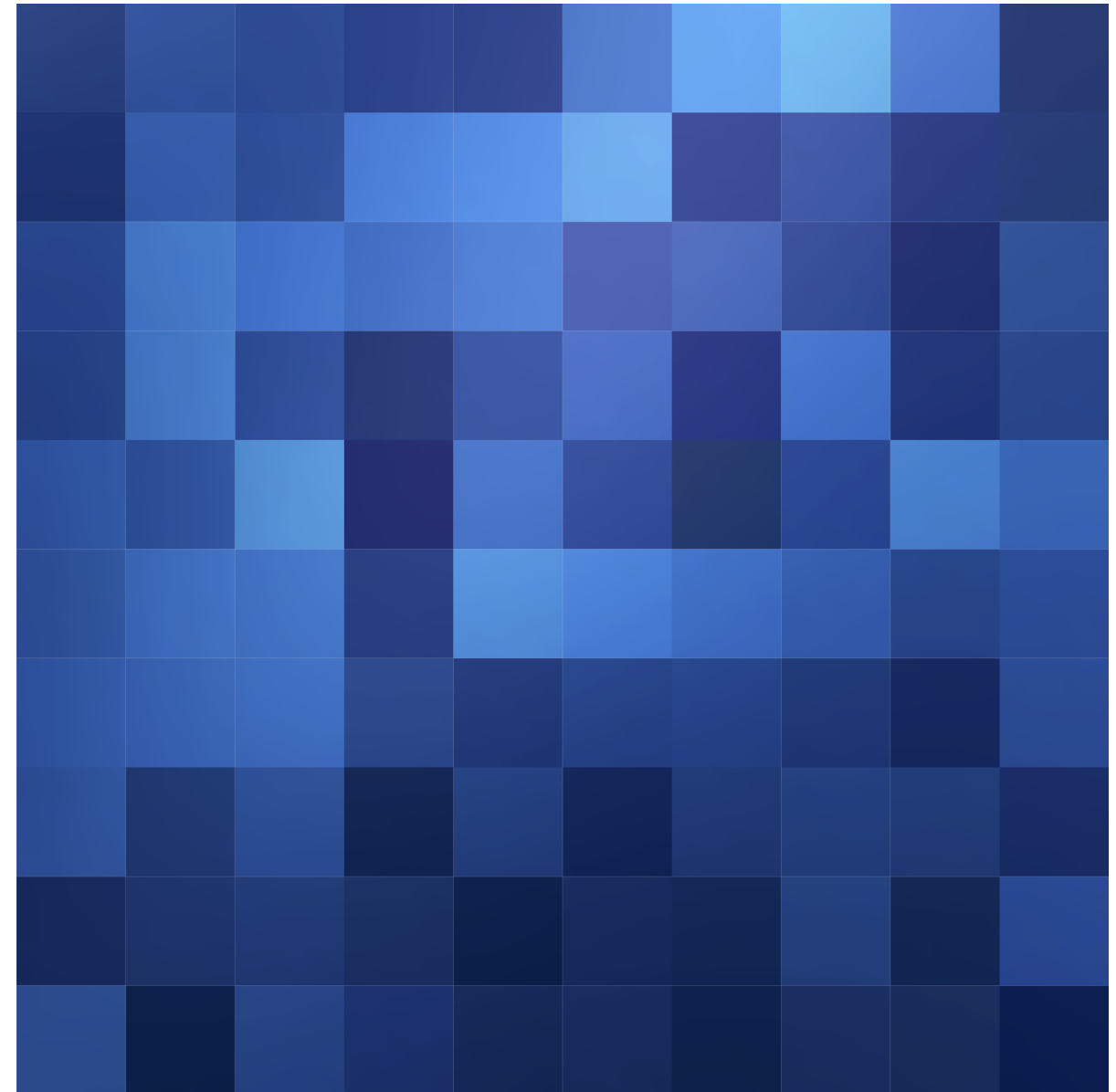
Sentence	Texans	Giants	football	great
The Texans are a great football team.	1	0	1	1
The Giants are a great football team.	0	1	1	1

Problems with word counts: negation

Sentence	Texans	Giants	football	great	not
The Giants are a great football team.	0	1	1	1	0
The Giants are not a great football team.	0	1	1	1	1

Word counts and synonyms

- Word counts don't help us consider synonyms
- Example: "blue"
 - "sky-blue"
 - "aqua"
 - "cerulean"
- Want to group as a single feature



Word embeddings

- Create features that group similar words
- Features have a mathematical meaning:

`king - man + woman = queen`

Review

- **Time series forecasting**
 - Time is a feature
 - Seasonality
- **Natural Language Processing (NLP)**
 - Text as input data
 - Word counts
 - Word embeddings

Let's practice!

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Deep Learning and Explainable AI

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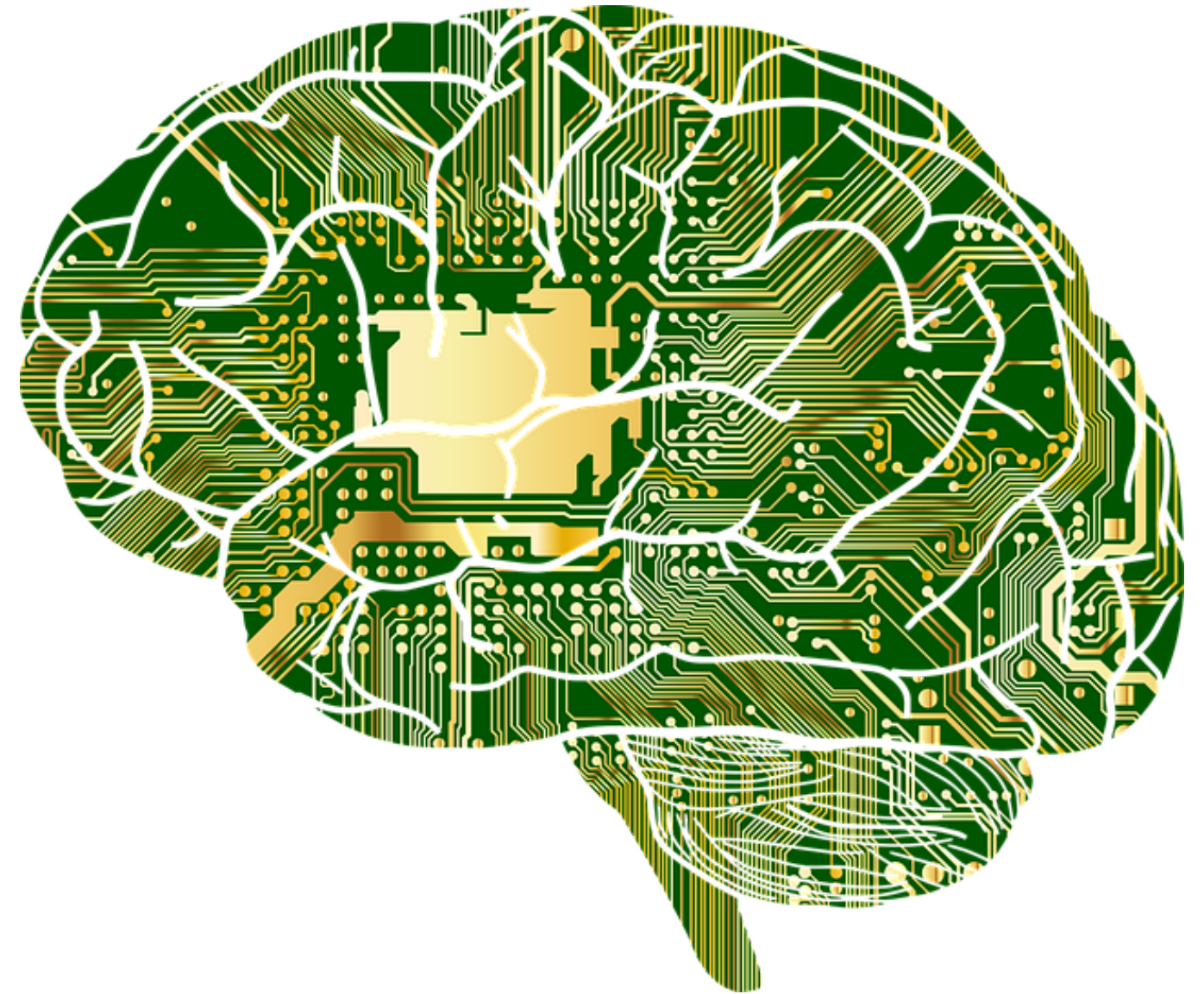


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What is Deep Learning?

- AKA: Neural Networks or Neural Nets
- Special area of Machine Learning
- Requires more data
- Best when inputs that are images or text



Explainable AI

Deep Learning	Explainable AI
Highly accurate predictions	Understandable by humans
Better for "What?"	Better for "Why?"

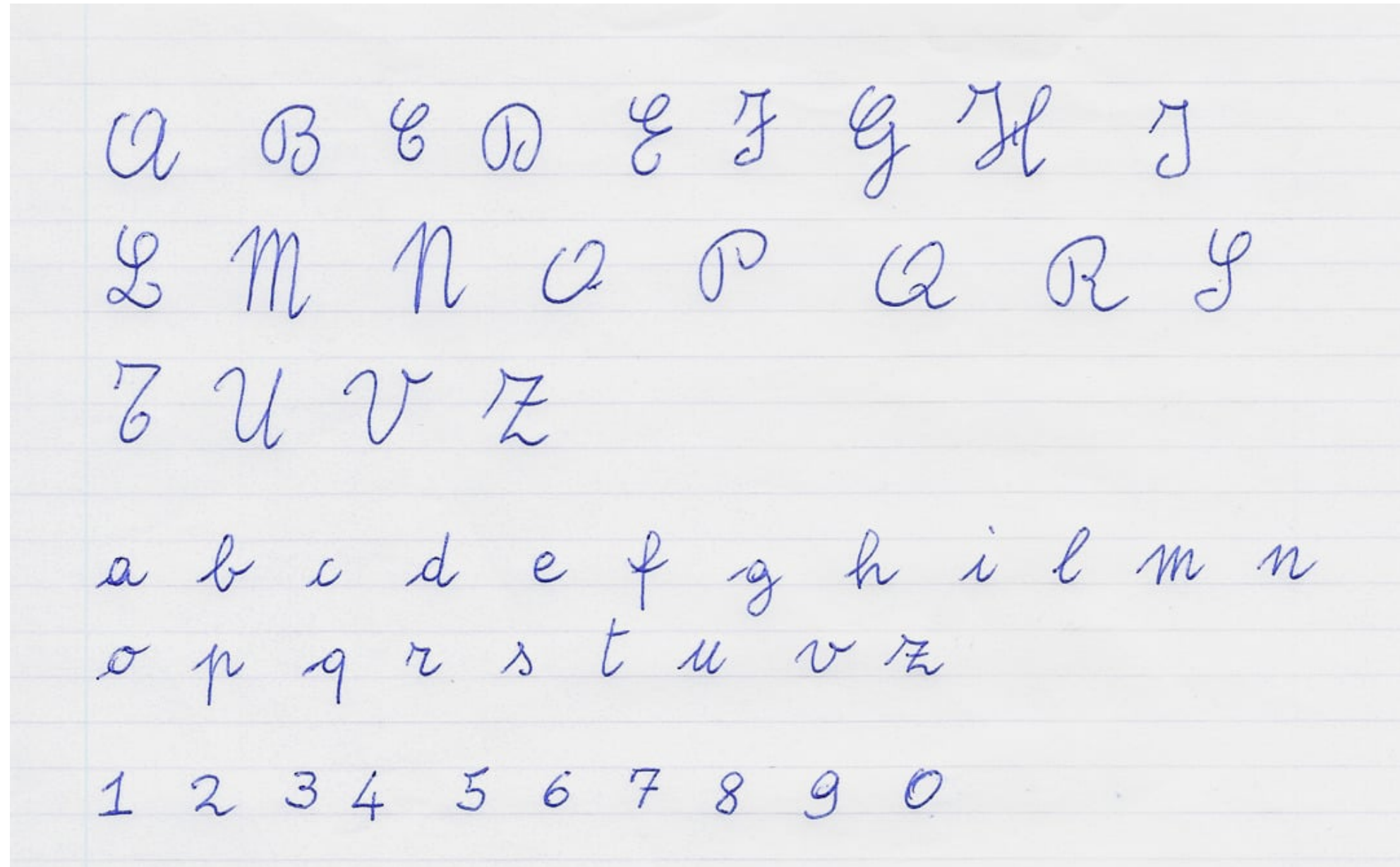
Case Study: Explainable AI

1. **Prediction:** What a customer is likely to do
2. **Explanation:** Why a customer is likely to do it



Case Study: Inexplicable AI

Prediction only: Which letter is this likely to be?



When to use Deep Learning

1. Is the training data complex?
2. Do we have a very large amount of training data?
3. Does the model need to be predictive or explanatory?

Let's practice!

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